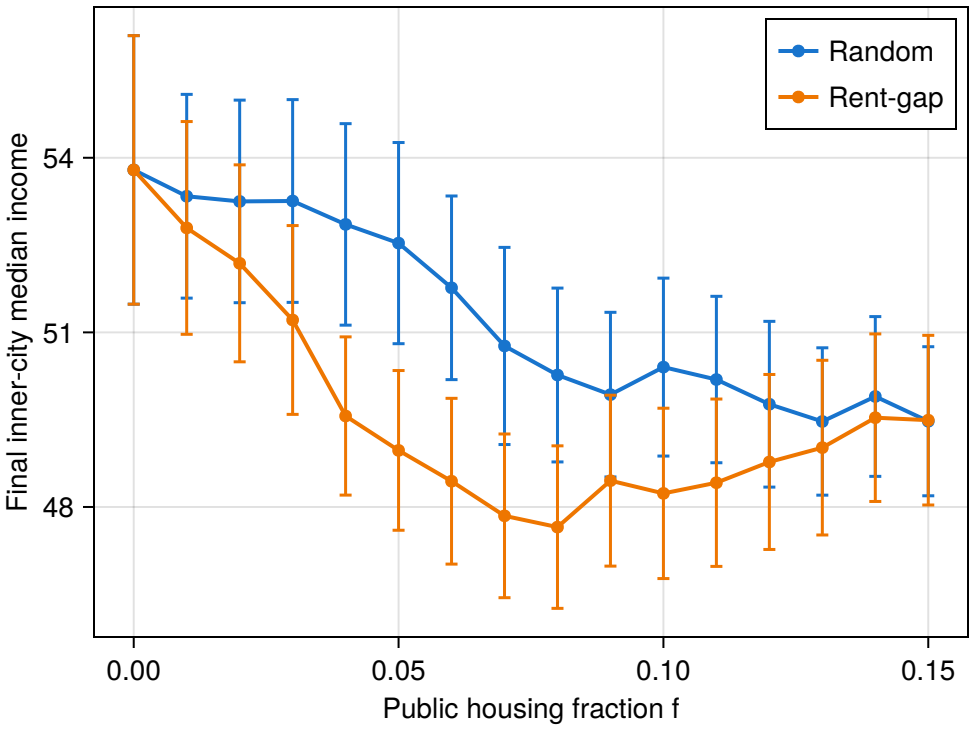
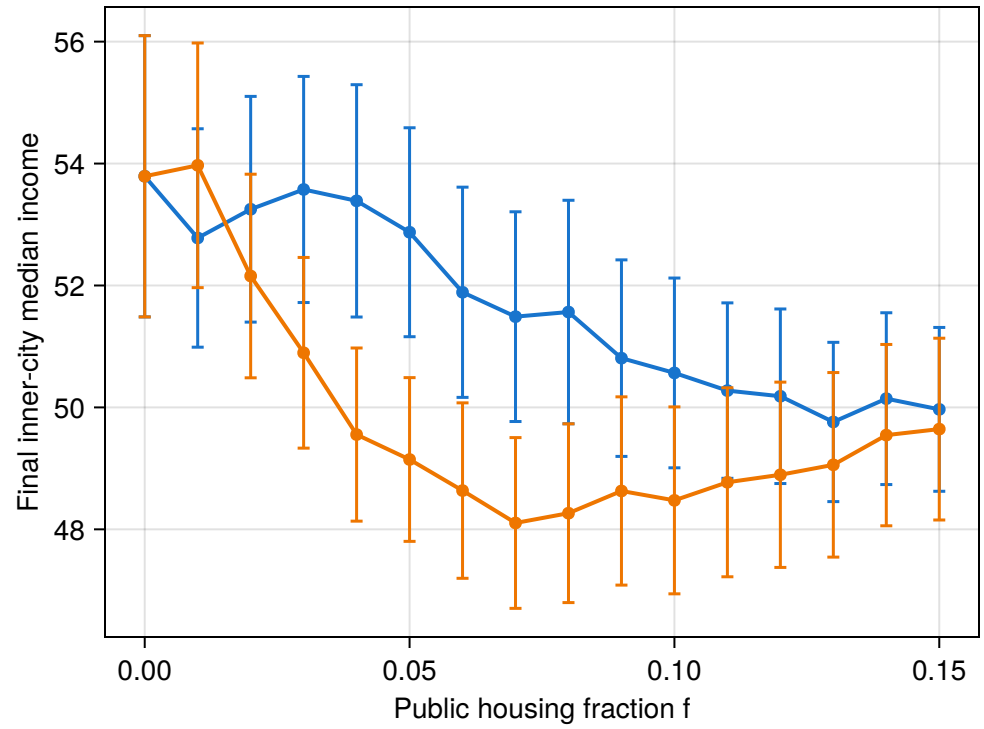
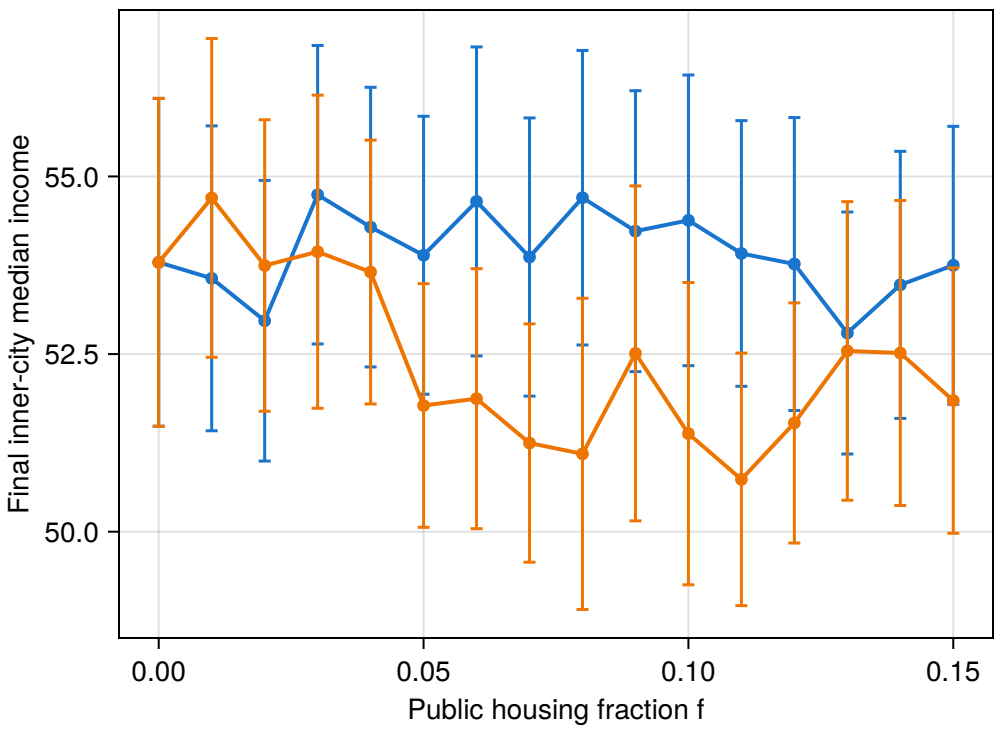
paper



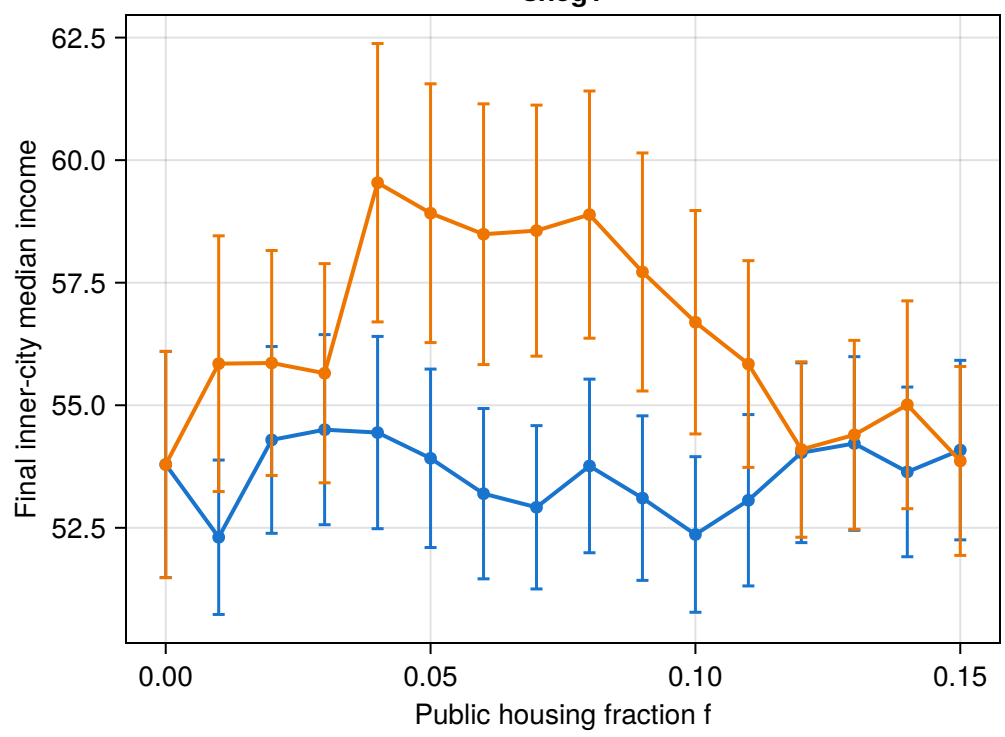
s1



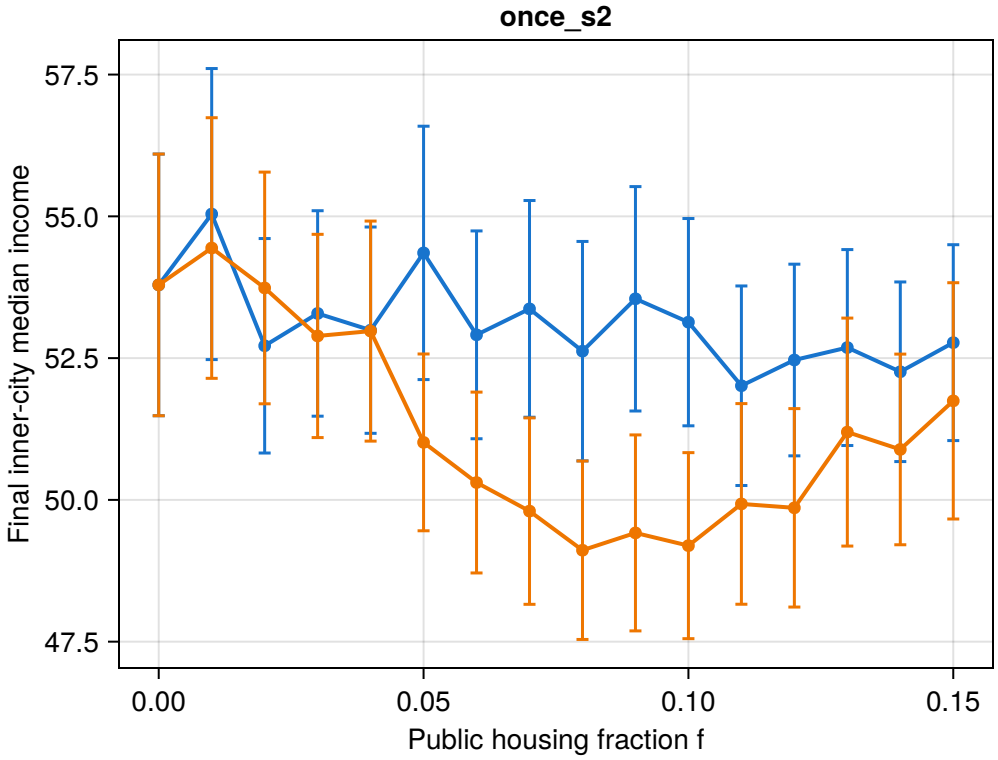
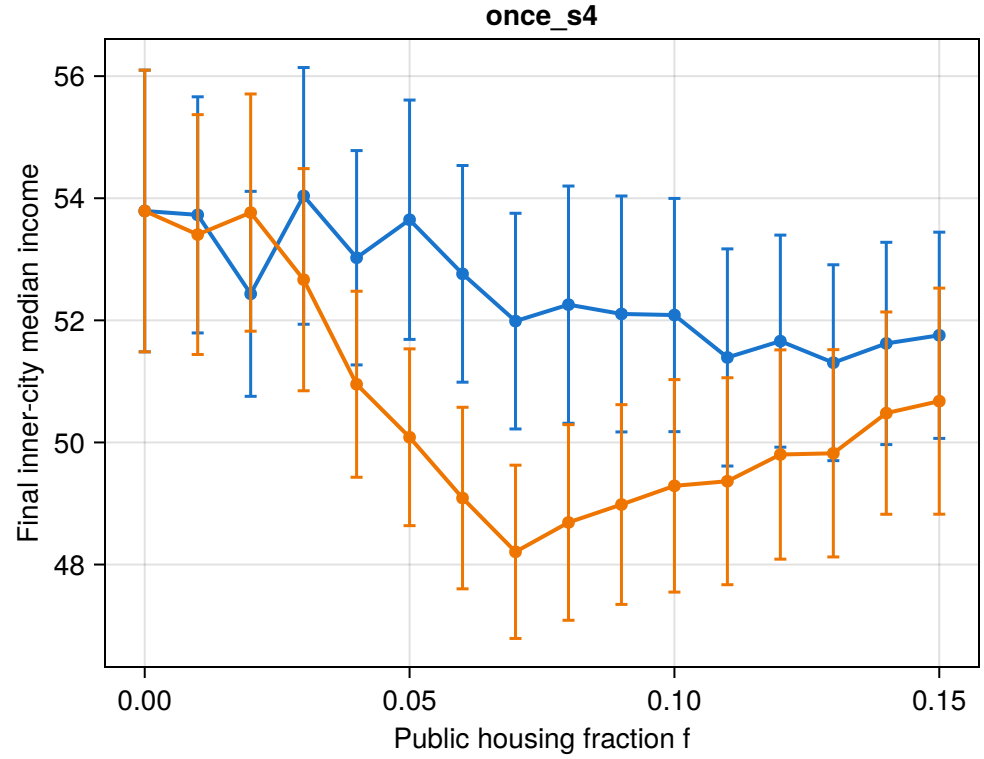
s0



sneg1



once_s2



once_s4